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Experimental and kinetic modeling study of the low- and intermediate-temperature oxidation of dimethyl ether

Zhandong Wang^{a,b,1}, Xiaoyuan Zhang^{b,1}, Lili Xing^b, Lidong Zhang^b, Friederike Herrmann^a, Kai Moshhammer^a, Fei Qi^{b,*}, Katharina Kohse-Höinghaus^{a,*}

^a Department of Chemistry, Bielefeld University, Universitätsstr. 25, D-33615 Bielefeld, Germany

^b National Synchrotron Radiation Laboratory, University of Science and Technology of China, Hefei, Anhui 230029, PR China

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ABSTRACT

Recent experiments on low-temperature oxidation of dimethyl ether (DME) at atmospheric pressure reveal much lower fuel conversion than the predictions of all published models. Consistent with previous measurements, the present work on DME oxidation in a laminar flow reactor between 400 and 1160 K at atmospheric pressure also confirmed this behavior. To reduce the gap between model predictions and experimental results, both regarding key oxygenated species and fuel conversion, an extended and updated kinetic model of DME oxidation was developed from the widely used model of Z. Zhao, M. Chaos, A. Kazakov, F.L. Dryer, *Int. J. Chem. Kinet.* 40 (2008) 1–18. The development of the model focused on the reaction sequences that affect the low-temperature oxidation reactivity. In particular, the reactivity of DME oxidation in the low-temperature regime and at atmospheric pressure can now be predicted. This is possible upon the inclusion of an additional, chemically-activated pathway of CH_3OCH_2 reaction with O_2 , as well as of the new decomposition chemistry of OCH_2OCHO radical, investigated theoretically in this work. The present model was examined against species concentrations in DME oxidation from low to intermediate temperatures and shock tube ignition delay times. It shows satisfactory performance in reproducing the respective literature data.

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1. Introduction

Combustion processes need to become cleaner and more efficient to meet stricter emission regulations and limited fossil fuel resources. An important attempt to address combustion energy challenges is the introduction of alternative fuels, especially bio-derived fuels. Generally, typical biofuels may reduce the emissions of polycyclic aromatic hydrocarbons (PAHs) and soot, while the formation of carbonyl compounds such as formaldehyde, acetaldehyde, acetone, and higher aldehydes and ketones often increases [1]. It is warranted to comprehensively study the combustion chemistries and emission properties of potential biofuels before their large-scale use in industry and transportation to examine whether they are environmentally friendly. An attractive renewable fuel is dimethyl ether (DME) which presents low soot and NO_x emissions and a high cetane number. It has thus been proposed as an alternative fuel and as an additive to diesel fuel

[2,3]. Its large-scale production from natural gas, biomass, and coal is possible, especially in China [4].

Due to its promising properties and potential as an alternative fuel, DME has been investigated extensively in the past decade including experiments such as pyrolysis [5–11], oxidation [4,5,8,12–24] and flames [25–36], and theoretical calculations [7–9,17–19,37–39], covering a wide range of temperatures and pressures. Based on these experimental and theoretical studies, detailed kinetic models for DME combustion have been developed [5,7,15,25,40]. In previous work, some experiments [13–16,21,22,24] addressed the low-temperature oxidation of DME. For example, earlier studies at relatively high pressure include experiments in a shock tube [13], jet-stirred reactor (JSR) [14] and flow reactor [15]. These measurements can be reproduced by previously published models [5,7,15,25,40]. However, in recent studies, Guo et al. [21] investigated DME oxidation from 490 to 750 K at atmospheric pressure in a flow reactor, and found that the model developed by Zhao et al. [7], named as “Zhao model” herein, predicted a much higher oxidation reactivity than their measurements. This observation was confirmed by Herrmann et al. [23] in their DME oxidation experiments with different equivalence ratios. More recently, Kurimoto et al. [24] also found discrepancies between

* Corresponding authors.

E-mail addresses: fqi@ustc.edu.cn (F. Qi), kkh@uni-bielefeld.de (K. Kohse-Höinghaus).

¹ Both have contributed equally to this work.

Table 1
Experimental conditions of the low-temperature oxidation of DME.

Oxidation	X_{DME}	X_{O_2}	X_{Ar}	P/atm	Total flow rate (SLM) ^a	Residence time (s)
Exp. 1	1.25	3.75	95	1	0.5	^b
Exp. 2	1.25	3.75	95	1	0.5	^b
Exp. 3	1.25	3.75	95	1	^c	1.5

Note: Exp. 1: flow reactor oxidation in Bielefeld; Exp. 2: flow reactor oxidation in Hefei; Exp. 3: JSR oxidation in Hefei.

^a Std. conditions: 273.15 K, 1 atm.

^{b,c} Change with temperature.

the predictions of the Zhao model [7] and their experimental results from a flow reactor. They improved the Zhao model to reproduce the measured fuel conversion by adjusting the branching ratio of the second oxygen addition channel and its competitive channel of the important QOOH intermediate to CH_2O . Moreover, some further advances of chemical kinetics and experimental diagnostics have offered useful insight regarding the key reactions [9,17–19,38,41] that affect the oxidation reactivity of DME in the low-temperature regime.

In this work, DME oxidation in a laminar flow reactor was investigated in the low- and intermediate-temperature regimes from 400 to 1160 K at atmospheric pressure. With a molecular-beam mass spectrometer, mole fraction profiles of reactants, intermediates and major products were measured. The modeling part started with the Zhao model [7] in which some key reactions and their related rate constants were updated with information from the above-mentioned recent experimental and theoretical investigations [9,17–19,38,41]. Furthermore, theoretical calculations were performed to improve this mechanism. In particular, our high-level quantum calculations on the decomposition chemistry of the OCH_2OCHO radical confirms a new formic acid pathway which was not included in previous models [5,7,15,25,40]. The present model shows good agreement with data in this work and further experimental results reported in the literature [13–15,21,22,24]. We believe that these results are helpful for the understanding of the low-temperature oxidation process of DME and may lead to further development of a comprehensive DME low-temperature oxidation mechanism in the future.

2. Experimental method

DME oxidation was investigated in detail using different experimental set-ups. One experiment was performed in a laminar flow reactor coupled to an electron ionization molecular-beam mass spectrometer (EI-MBMS) at Bielefeld University. The other two

experiments, a laminar flow reactor [42,43] and a JSR [44], were coupled to photoionization molecular-beam mass spectrometry (PI-MBMS) at the National Synchrotron Radiation Laboratory in Hefei, China. The measurement in Hefei was focused on the investigation of the low-temperature regime oxidation and the identification of the species via photoionization efficiency (PIE) spectra, to support the results of the Bielefeld flow reactor experiment. Thus, only the Bielefeld set-up is described here. Information on the Hefei set-ups can be found in the [Supplementary Material](#). The experimental conditions of the three experiments are shown in Table 1.

As a continuous development of our previous studies [22,23], a new apparatus for the analysis of low-temperature oxidations was designed in the present work. The laminar flow reactor used in this study consisted of a fused silica tube with an inner diameter of 8 mm and a total length of 830 mm. As shown in Fig. 1, the reactor was directly coupled to the time-of-flight mass spectrometer as a molecular beam experiment. The used flange system (Fig. 1, right panel) was important to optimize the position of the quartz nozzle inside the reactor tube. The exhaust gas was pumped via four outlet tubes that are positioned symmetrically enabling undisturbed sampling. The reactor tube can be heated by three independent heating sections. The length of the three heating zones (T1, T2, and T3) is approximately 300, 200, and 200 mm, respectively. Each part was monitored by a Ni–Cr/Ni thermocouple. It should be noticed that the wall reactions in the present quartz flow tube are expected to be negligible due to the small surface area to volume ratio of 5 cm^{-1} , compared with previous studies [5,15,16,21,45,46], and we did not see any indications of surface reactions.

As shown in Fig. 1, the electron-ionization time-of-flight mass spectrometer (Kaesdorf) was directly adapted to the laminar flow reactor. The gas was sampled via a quartz nozzle ($\sim 50\text{ }\mu\text{m}$ orifice diameter) and expanded into the 1st pumping stage (Fig. 1, left panel) pumped down to 10^{-4} mbar by a turbo-molecular pump. The formed molecular beam was further sampled by a copper skimmer at the 2nd pumping stage, and then extracted to the

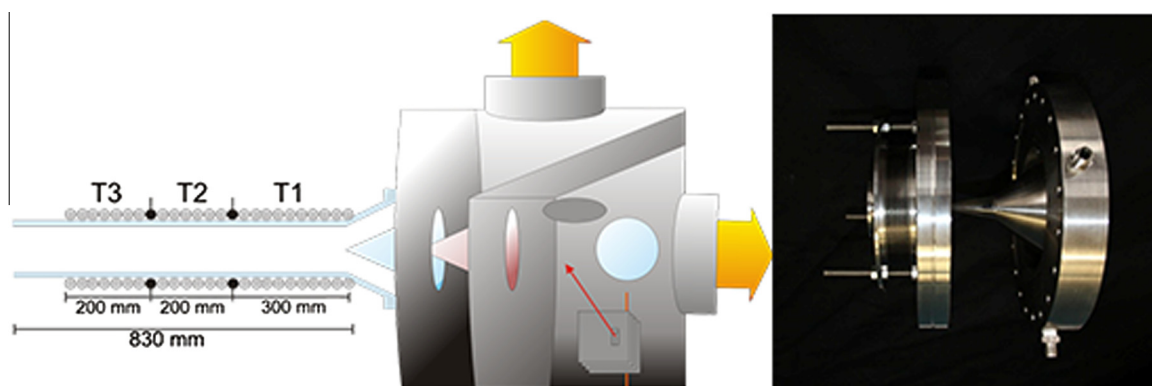


Fig. 1. Schematic diagram of the laminar flow reactor coupled with a molecular-beam mass spectrometer (left panel); flange system connected to the flow reactor and the MBMS system (right panel).

Table 2The updated reactions and the rate coefficients of DME low-temperature oxidation in this work (units: $\text{cm}^3/\text{s}/\text{mol}/\text{cal}$).

No.	Reactions	A	n	E	P (atm)	Refs.
R1	$\text{CH}_3\text{OCH}_3 + \text{OH} = \text{CH}_3\text{OCH}_2 + \text{H}_2\text{O}$	1.95E+07	1.89	−366		[41]
R2	$\text{CH}_3\text{OCH}_3 + \text{H} = \text{CH}_3\text{OCH}_2 + \text{H}_2$	3.94E+00	4.13	1780		[9]
R3	$\text{CH}_3\text{OCH}_3 (+\text{M}) = \text{CH}_3\text{O} + \text{CH}_3 (+\text{M})$	2.33E+19	−0.66	84,139		[9]
	Low pressure limit	1.72E+59	−11.40	93,296		
	Troe: 1.0 1.0E−30 8.8E+2					
R4	$\text{CH}_3\text{OCH}_2 + \text{O}_2 = \text{CH}_3\text{OCH}_2\text{OO}$	4.11E+19	−2.45	1137	1	[19]
		1.12E+14	−0.58	−164	10	[19]
		1.87E+12	0.00	−648	HPL	[19]
R5	$\text{CH}_3\text{OCH}_2 + \text{O}_2 = \text{OH} + \text{CH}_2\text{O} + \text{CH}_2\text{O}$	3.50E+17	−1.45	4718	1	[19]
		9.53E+10	0.42	3416	10	[19]
		1.59E−05	1.00	2932	HPL	[19]
R6	$\text{CH}_3\text{OCH}_2 + \text{O}_2 = \text{HCO} + \text{H}_2\text{O} + \text{CH}_2\text{O}$	6.67E+10	0.00	773		[18]
R7	$\text{CH}_3\text{OCH}_2\text{OO} = \text{CH}_2\text{OCH}_2\text{OOH}$	2.20E+09	0.00	15,846		[17]
R8	$2\text{CH}_3\text{OCH}_2\text{OO} = \text{CH}_3\text{OCH}_2\text{O} + \text{CH}_3\text{OCH}_2\text{O} + \text{O}_2$	1.26E+11	0.00	−1390		[19] ^a
R9	$2\text{CH}_3\text{OCH}_2\text{OO} = \text{CH}_3\text{OCHO} + \text{CH}_3\text{OCH}_2\text{OH} + \text{O}_2$	5.42E+10	0.00	−1390		[19] ^a
R10	$\text{CH}_2\text{OCH}_2\text{OOH} + \text{O}_2 = \text{OOCH}_2\text{OCH}_2\text{OOH}$	7.00E+11	0.00	0		[7]
R11	$\text{CH}_2\text{OCH}_2\text{OOH} = \text{OH} + \text{CH}_2\text{O} + \text{CH}_2\text{O}$	1.50E+13	0.00	20,500		[7]
R12	$\text{OOCH}_2\text{OCH}_2\text{OOH} = \text{HOOCH}_2\text{OCHO} + \text{OH}$	4.00E+10	0.00	18,500		[7]
R13	$\text{HOOCH}_2\text{OCHO} = \text{OCH}_2\text{OCHO} + \text{OH}$	3.00E+16	0.00	40,000		[7]
R14	$\text{OCH}_2\text{OCHO} \Rightarrow \text{HOCHO} + \text{HCO}$	5.17E+20	−3.78	11,396	1	This work
		3.18E+25	−4.88	14,509	10	This work
		1.10E+10	0.72	13,200	HPL	This work
R15	$\text{OCH}_2\text{OCHO} \Rightarrow \text{HOCH}_2\text{OCO}$	4.19E+52	−15.11	21,607	1	This work
		4.82E+56	−15.73	24,786	10	This work
		3.33E+08	1.08	15,714	HPL	This work
R16	$\text{OCH}_2\text{OCHO} \Rightarrow \text{OCHOCHO} + \text{H}$	2.39E+22	−5.25	14,385	1	This work
		3.32E+28	−6.27	18,481	10	This work
		1.87E+09	1.52	19,887	HPL	This work
R17	$\text{OCH}_2\text{OCHO} \Rightarrow \text{CH}_2\text{OH} + \text{CO}_2$	2.22E+19	−5.01	14,057	1	This work
		9.96E+25	−6.28	18,454	10	This work
R18	$\text{OCH}_2\text{OCHO} \Rightarrow \text{HOCH}_2\text{O} + \text{CO}$	9.26E+20	−4.46	12,921	1	This work
		4.89E+26	−5.67	16,935	10	This work
−R15	$\text{HOCH}_2\text{OCO} \Rightarrow \text{OCH}_2\text{OCHO}$	7.54E+23	−6.18	17,437	1	This work
		1.70E+30	−7.54	21,267	10	This work
R19	$\text{HOCH}_2\text{OCO} \Rightarrow \text{H} + \text{OCHOCHO}$	1.26E+30	−8.71	23,542	1	This work
		2.13E+38	−10.18	28,794	10	This work
R20	$\text{HOCH}_2\text{OCO} \Rightarrow \text{HCO} + \text{HOCHO}$	9.20E+29	−7.74	21,725	1	This work
		2.84E+37	−9.45	26,661	10	This work
R21	$\text{HOCH}_2\text{OCO} \Rightarrow \text{CH}_2\text{OH} + \text{CO}_2$	2.41E+29	−7.69	21,150	1	This work
		9.11E+34	−8.64	24,627	10	This work
		5.06E+12	0.22	25,210	HPL	This work
R22	$\text{HOCH}_2\text{OCO} \Rightarrow \text{HOCH}_2\text{O} + \text{CO}$	2.53E+39	−9.30	22,931	1	This work
		5.70E+35	−7.82	22,449	10	This work
		7.17E+13	0.16	20,606	HPL	This work
R23	$\text{CH}_2\text{O} + \text{OH} = \text{HCO} + \text{H}_2\text{O}$	1.94E+06	2.11	−1690		[65]
R24	$\text{CH}_2\text{O} + \text{OH} = \text{HOCH}_2\text{O}$	6.31E+06	1.63	4280		[65]
R25	$\text{H} + \text{O}_2 (+\text{M}) = \text{HO}_2 (+\text{M})$	4.65E+12	0.44	0		[40]
	Low pressure limit	1.74E+19	−1.23	0		
	Troe: 6.7E−1 1.0E−30 1.0E+30 1.0E+30					
	$\text{H} + \text{O}_2 (+\text{AR}) = \text{HO}_2 (+\text{AR})$	4.65E+12	0.44	0		[40]
	Low pressure limit	6.81E+18	−1.20	0		
	Troe: 7.0E−1 1.0E−30 1.0E+30 1.0E+30					
	$\text{H} + \text{O}_2 (+\text{HE}) = \text{HO}_2 (+\text{HE})$	4.65E+12	0.44	0		[40]
	Low pressure limit	9.19E+18	−1.20	0		
	Troe: 5.9E−1 1.0E−30 1.0E+30 1.0E+30					

Note: HPL: high pressure limit.

^a Branching ratio estimated from Jenkin et al. [63].

ionization chamber ($<10^{-6}$ mbar). The distance between the skimmer and nozzle was ~ 20 mm. The species were ionized with an electron beam and then accelerated into the reflectron time-of-flight mass spectrometer separating the ions by their different masses. For the detection, dual multichannel plates were used. The mass resolution of this mass spectrometer ($m/\Delta m$) was ~ 2300 . The measurements were performed at an electron energy of 20 eV.

The premixed stoichiometric gas mixture containing 1.25% DME, 3.75% O_2 , and 95% Ar was fed into the reactor inlet. Self-sustaining combustion was suppressed because of the high dilution. The flow rates of the gases were regulated by calibrated mass flow controllers, and the total flow rate was 0.5 standard liters per minute (SLM) (std. conditions: 273.15 K, 1 atm). The gas composition

was determined via quartz nozzle sampling as a function of the reaction temperature by using ramps of 0.05 K/s from 1160 K to 400 K. Furthermore, temperature profiles along the reactor tube were measured at a total argon flow of 0.5 SLM by moving a Ni–Cr/Ni thermocouple along the centerline of the reactor. Figure 2 shows four temperature profiles measured at isothermal conditions as a function of the reactor length L , and each temperature profile is identified by its measured maximum value T_{max} . The gas inlet is at $L = 0$ mm and the sampling nozzle at $L = 830$ mm. The small temperature increase at $L = 560$ mm is caused by an inhomogeneity of heat conduction. However, due to small inhomogeneity the absolute temperature error is below 5% and the thermocouples exhibit an uncertainty of less than 0.4%. The mole fraction evaluation used in this study is adopted from

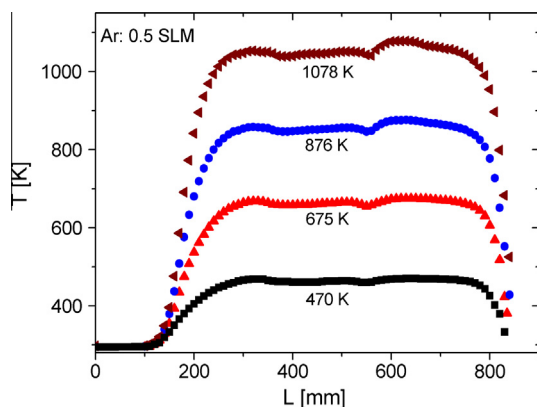


Fig. 2. Centerline temperature profiles measured under isothermal conditions for a flow of 0.5 SLM Ar; maximum temperatures are indicated.

[22,47–49]. Generally, the uncertainties of the reactants are estimated to be within $\pm 10\%$, and those for the major species are estimated to be within $\pm 30\%$. Calibrations for intermediates that use the relative ionization cross section (RICS) method [50] or the convolution of literature cross sections with the known energy distribution of the ionizing electrons are typically within a factor of 2–4.

3. Theoretical calculations

3.1. Calculation method

In this work, a detailed investigation was performed to identify the reaction pathways of OCH_2OCHO radical, an important intermediate formed from the oxygen addition and chain branching process of DME at low-temperature oxidation conditions:



(Numbers for all reactions discussed here refer to Table 2.)

The unimolecular decomposition pathways of OCH_2OCHO were first optimized at the B3LYP/6-311++G(d,p) level. Then the high

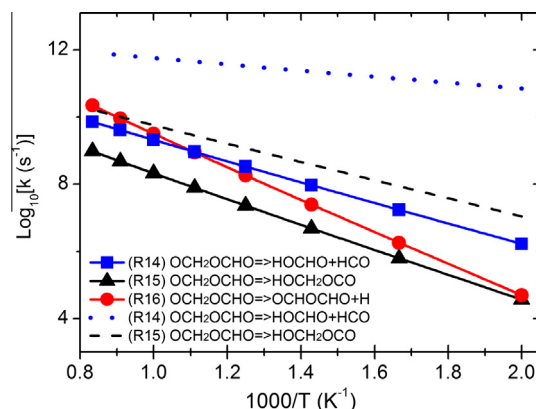


Fig. 4. High-pressure-limit rate constants of the direct dissociation (R14) and (R16) and isomerization (R15) of OCH_2OCHO calculated by MESMER [60,61] at 500–1200 K in this work (symbols plus solid lines), also with rate constants of (R14) and (R15) calculated by Andersen and Carter [38] (dash and dot lines).

precision single-point energies were determined with the method of quadratic configuration interaction with singles, doubles and perturbative inclusion of triples (UQCISD(T)) [51,52] with two basis sets: the correlation-consistent, polarized-valence, triple- ζ (cc-PVTZ), and quadruples- ζ (cc-PVQZ) basis sets [52,53]. The UQCISD(T) energies were extrapolated to the complete basis set limit (CBS) via the following expression [53]:

$$E[\text{UQCISD(T)}/\infty] \approx E[\text{UQCISD(T)}/\text{cc-PVQZ}] + \{E[\text{UQCISD(T)}/\text{cc-PVQZ}] - E[\text{UQCISD(T)}/\text{cc-PVTZ}]\} * 0.6938. \quad (\text{E1})$$

The zero-point energy corrections were obtained from the B3LYP/6-311++G(d,p) optimizations. All calculations were performed with the Gaussian09 program [54]. The present calculations at the level of UQCISD(T)/CBS//B3LYP/6-311++G(d,p) have

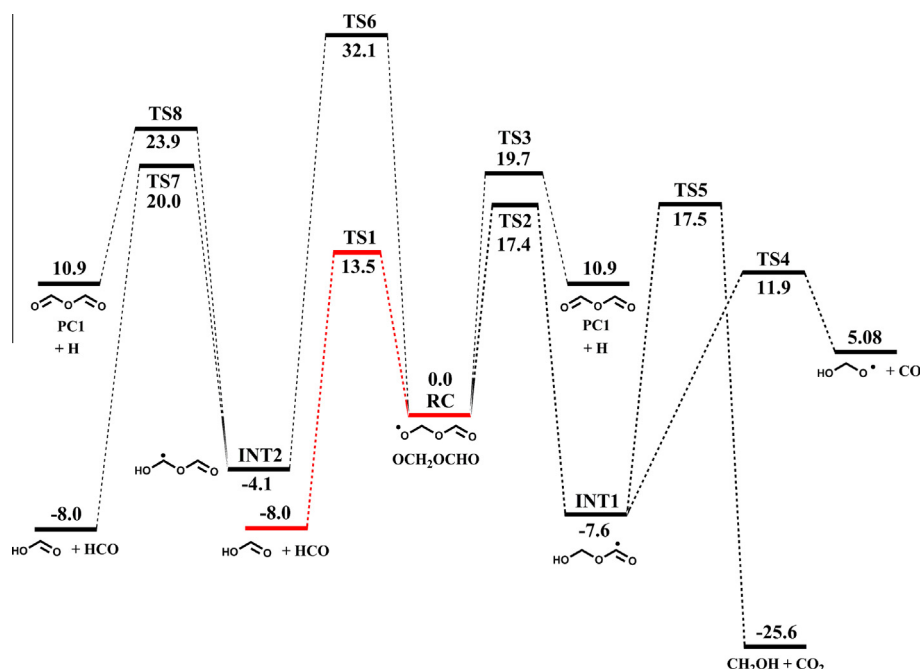


Fig. 3. The potential energy surfaces for OCH_2OCHO decomposition and isomerization performed at the UQCISD(T)/CBS//B3LYP/6-311++G(d,p) level. The unit for energies is kcal/mol.

an uncertainty of 1 kcal/mol according to the evaluation by Goldsmith et al. [55].

To assess the status and branching ratios of OCH_2OCHO pathways in combustion systems, especially in the low-temperature oxidation regimes, rate constant calculation was also performed in this work. The vibrational frequencies and rotational constants

obtained from the B3LYP/6-311++G(d,p) method were adopted in the following rate constant calculations. The temperature- and pressure-dependent phenomenological rate constants for the main decomposition channels were determined with the RRKM/Master Equation method. The collision energy transfer adopted in the collisional model was treated using a single-parameter exponential

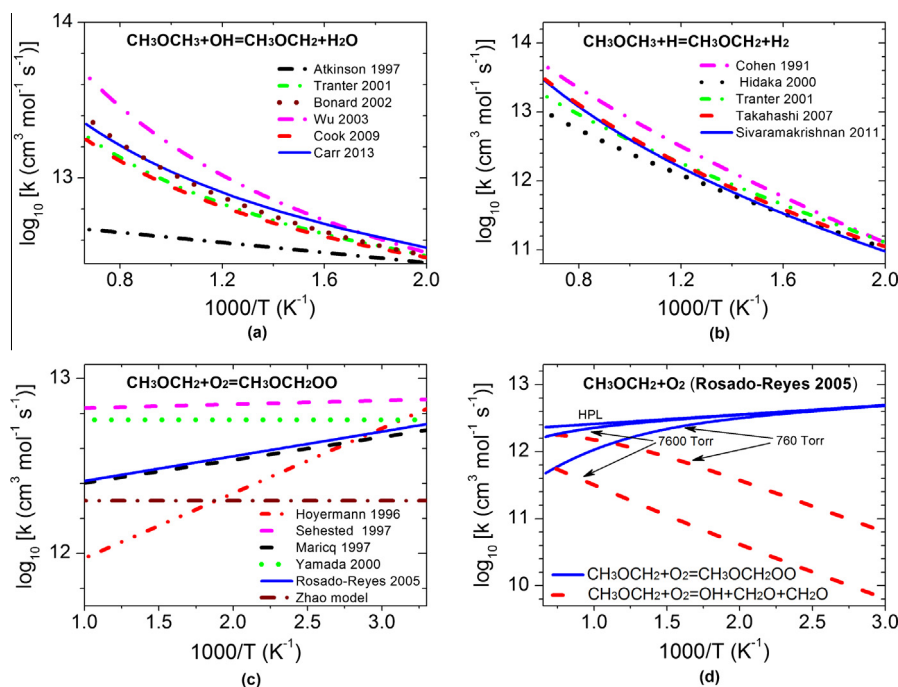


Fig. 5. Comparison of the rate constants of H-abstraction reactions via (a) the OH radical by Atkinson et al. [66], Tranter and Walker [67], Bonard et al. [68], Wu et al. [69], Cook et al. [8], and Carr et al. [41], as well as via (b) the H atom by Cohen [70], Hidaka et al. [6], Tranter and Walker [67], Takahashi et al. [71], and Sivaramakrishnan et al. [9]; comparison of the high-pressure-limit rate constants of (c) $\text{CH}_3\text{OCH}_2 + \text{O}_2 = \text{CH}_3\text{OCH}_2\text{OO}$ by Hoyermann and Nacke [72], Sehested et al. [73], Maricq et al. [62], Yamada et al. [39], Rosado-Reyes et al. [19], and the Zhao model [7], as well as pressure-dependent rate constant of (d) $\text{CH}_3\text{OCH}_2 + \text{O}_2$ reactions by Rosado-Reyes et al. [19].

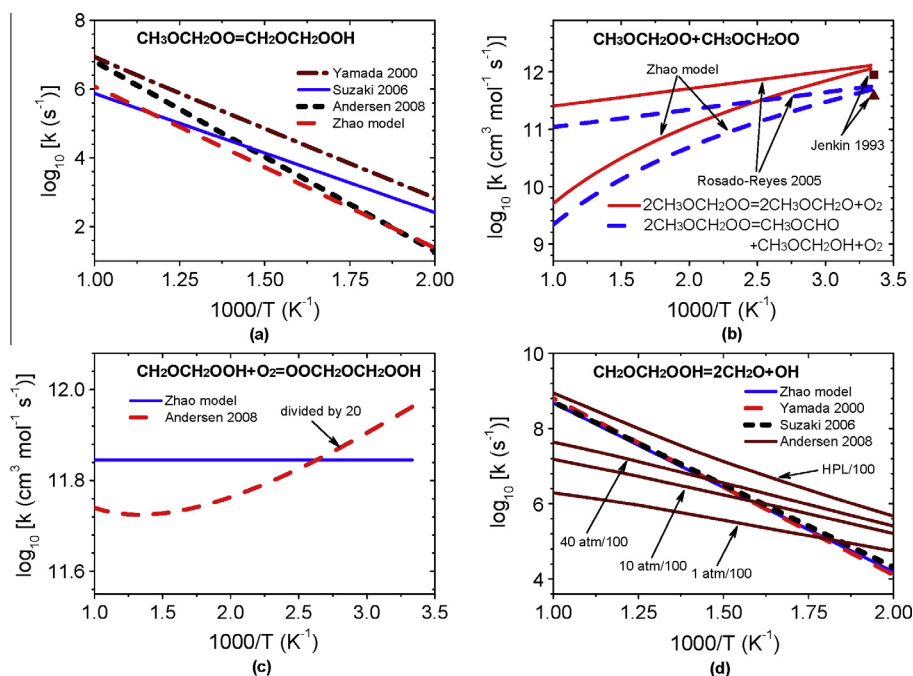


Fig. 6. Comparison of the high-pressure-limit rate constants of (a) isomerization of $\text{CH}_3\text{OCH}_2\text{OO}$ by Yamada et al. [39], Andersen and Carter [38], Suzuki et al. [17], and the Zhao model [7]; (b) the self-reaction of $\text{CH}_3\text{OCH}_2\text{OO}$ by Rosado-Reyes et al. [19], the Zhao model [7], and Jenkin et al. [63] at 298 K (solid square: R8; solid triangle: R9); (c) $\text{CH}_2\text{OCH}_2\text{OOH} + \text{O}_2 = \text{OCH}_2\text{OCH}_2\text{OOH}$ by Andersen and Carter [38], and the Zhao model [7], as well as (d) $\text{CH}_2\text{OCH}_2\text{OOH} = \text{OH} + \text{CH}_2\text{O} + \text{CH}_2\text{O}$ by Yamada et al. [39], Suzuki et al. [17], and the Zhao model [7]. The values of (c) and (d) by Andersen and Carter [38] are divided by a factor of 20 and 100 for a better comparison.

down model: $\langle \Delta E \rangle_{\text{down}} = 100 \times (T/300)^{0.85} \text{ cm}^{-1}$, following a previous study [41]. Moreover, the interaction between reactants and the bath gas Ar was modeled by the Lennard–Jones (L–J) potential. The L–J parameters were chosen as $\sigma = 3.465 \text{ \AA}$ and $\varepsilon = 113.5 \text{ K}$ for Ar [56]. For OCH_2OCHO , the L–J parameters were estimated according to the empirical equations [57]:

$$\sigma = 2.44(T_c/P_c)^{1/3}, \quad (\text{E2})$$

$$\varepsilon_A/k_b = 0.77T_c. \quad (\text{E3})$$

Here, k_b is the Boltzmann constant, T_c is the critical temperature and P_c is the critical pressure; the method of Constantinou and Gani [58] was used to estimate the values of T_c and P_c . According to equations E2 and E3, the L–J values for OCH_2OCHO were calculated to be $\sigma = 5.120 \text{ \AA}$ and $\varepsilon = 446.21 \text{ K}$. In the rate constant calculations, special consideration was given to the low-frequency vibrational modes. These modes were treated as hindered rotors [59] by B3LYP/6-311++G(d,p) hindrance potentials. The hindrance potentials were obtained by relaxed scanning at every 10 degrees, the results of which were fitted by a Fourier series up to 14 terms. Reactions were treated with the conventional transition state theory, while the number of states was evaluated by the rigid rotor harmonic-oscillator (RRHO) assumption. Furthermore, partition

functions were evaluated at the level of energy-resolved calculation and the grain size was chosen to be 40 cm^{-1} . The pressure covers the range of 1–10 atm, whereas the temperature varies from 500 to 1200 K. The phenomenological rate constants were calculated using the MESMER program [60,61].

3.2. Calculation results

The potential energy surfaces of OCH_2OCHO (RC) decomposition at UQCISD(T)/CBS//B3LYP/6-311++G(d,p) are shown in Fig. 3. The detailed geometries, energies and vibrational frequencies are provided in the Supplementary Material. The channel with the lowest energy barrier is the one via transition state (TS1) to form formic acid and HCO (R14, see below). Our calculation shows that the isomerization channel (R15) of OCH_2OCHO via a five-member-ring transition state (TS2), which was considered as the dominant channel of OCH_2OCHO decomposition in previous models [5,7,15,25,40], has an energy barrier of 17.4 kcal/mol, which is approximately 4 kcal/mol higher than TS1. Therefore, the reaction pathway calculation implies that R14 via TS1 is more readily to happen than R15 via TS2. This preliminary conclusion is in accord with the theoretical calculation by Andersen and Carter [37,38]. However, the energy barrier difference of the two channels calculated by Andersen and Carter (8 kcal/mol) is larger than that in this work (4 kcal/mol),

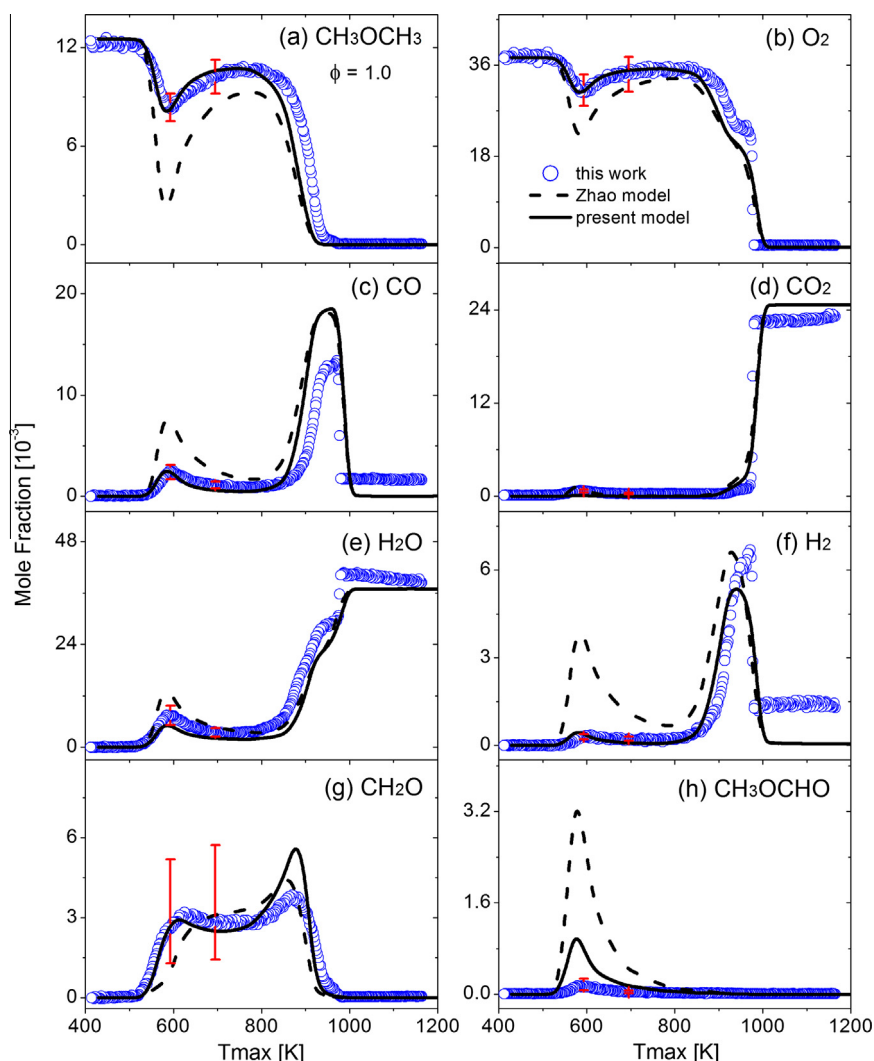


Fig. 7. Mole fraction profiles of selected species as a function of temperature in stoichiometric DME oxidation. Symbols are experimental results from this work; lines are simulation results. Dashed lines: Zhao model [7]; solid lines: present model.

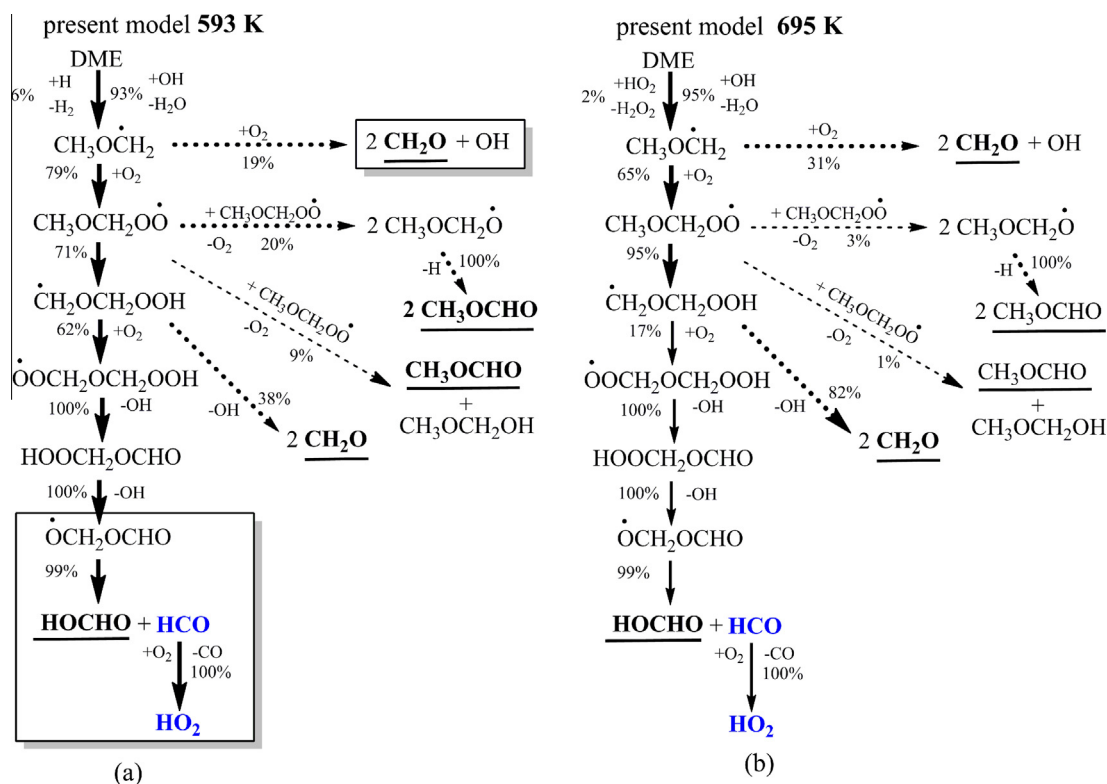


Fig. 8. Reaction flux analysis for DME consumption at 593 and 695 K from the present model. The major differences with the Zhao model [7] at 593 K are highlighted in the solid boxes.

due to different calculation methods used by Andersen and Carter and in this work, which will be discussed in detail below.



Apart from the channels discussed above, two other pathways exist for OCH_2OCHO decomposition, i.e. the β -C-H scission (R16) to form $OCHOCHO$ (PC1 in Fig. 3) via transition state (TS3), and the isomerization through a three-membered ring (TS6) to form $HOCHOCHO$. From the viewpoint of the energy barrier, the pathway via TS6 is not important, while the β -C-H scission (R16) should not be ignored. In addition, an intrinsic reaction coordinate scan for a 5-center transition state leading from OCH_2OCHO (RC) to $HOCHOCHO$ (INT2) was attempted but this did not converge instead of the direct dissociation pathway leading to $HCO + HOCHO$ products; the other reaction pathway to form $CH_2O + OCHO$ products, which was proposed by Curran et al. [15], shows a barrierless dissociation channel and the contribution to the overall OCH_2OCHO dissociation can be neglected based on our preliminary kinetic calculations.

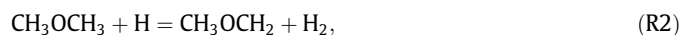
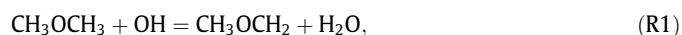
Figure 4 presents the high-pressure-limit (HPL) results of R14, R15 and R16 calculated by MESMER [60,61], which shows that the pathway R14 is predominant in the 500–800 K region. It is also noticeable that R16 begins to be competitive with R14 above around 830 K. Comparing with R14, the isomerization channel R15 is not important. The pressure- and temperature-dependent rate constants of the reaction pathways of OCH_2OCHO (R14–R22) calculated in this work are presented in Table 2. The calculated HPL rate constants of R14 and R15 by Andersen and Carter [38] are also given in Fig. 4. The results by Andersen and Carter also show that the OCH_2OCHO dissociation channel R14 is dominant

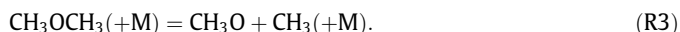
for 500–1200 K. However, the absolute rate constants and branching ratios of these two reactions are quite different from the result in this work. For example, the rate constants of R14 and R15 calculated in this work are 1.7×10^7 and $6.2 \times 10^5 \text{ s}^{-1}$ at 600 K, respectively, contrasted with those for R14 and R15 by Andersen and Carter [38] which are 1.4×10^{11} and $8.8 \times 10^7 \text{ s}^{-1}$ at 600 K, respectively. The branching ratio of R14 to R15 is 27 in this work while it is 1590 in that of Andersen and Carter. The large discrepancies between their work and ours mainly lie in their lower-level calculations at the B3LYP//6-311G** level, which have an energy error of 3–4 kcal/mol [38]. For OCH_2OCHO reactions, our calculated reaction pathways and the pressure-dependent rate constants were used in the updated model.

4. Kinetic modeling

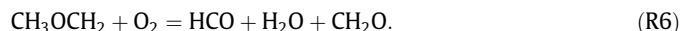
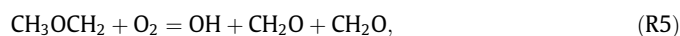
4.1. Modeling construction

The present model is based on the Zhao model [7]. We updated the pathways and rate constants of key reactions after comprehensive review and careful evaluation with respect to the recent experimental and theoretical investigations. The first targets are the H-abstraction reactions of DME by the OH radical (R1) and by the H atom (R2), which are dominant channels consuming DME. The rate comparisons of R1 and R2 from the literature are shown in Fig. 5a and b, respectively. In this work, we used the rate constants of R1 and R2 from the recent work of Carr et al. [41] and Sivaramakrishnan et al. [9]. Furthermore, the unimolecular decomposition of DME (R3) was also modified from the recent evaluation of Sivaramakrishnan et al. [9] in the Troe falloff format:

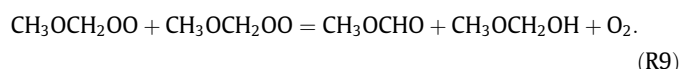
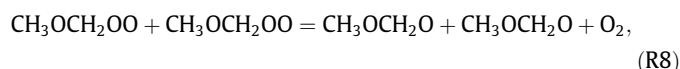




The methoxymethyl radical (CH_3OCH_2) is formed from H-abstraction reactions, then reacts with O_2 producing an excited methoxymethyl peroxy radical ($\text{CH}_3\text{OCH}_2\text{OO}^*$). It can undergo collisional relaxation or chemically-activated dissociation reactions depending upon the pressure and temperature. In the Zhao model, however, the rate constant for R4 is at the high-pressure-limit condition, which is hardly sufficient to predict the combustion behavior at various pressures. In the updated sub-mechanism of DME, rate constants of R4 and an additional chemically-activated reaction R5 evaluated by Rosado-Reyes et al. [19] were chosen. They measured the two product channels by using flash photolysis/transient infrared spectroscopy. Figure 5c presents the literature HPL rate constants of CH_3OCH_2 with O_2 , where $\text{CH}_3\text{OCH}_2\text{OO}$ is formed (R4). Large discrepancies exist among the literature values, and the selected rate constant from Rosado-Reyes et al. in the present work is close to the measurements from Maricq et al. [62], both of which are in the middle region among the literature values. Figure 5d shows the rate constants of R4 and R5 investigated by Rosado-Reyes et al. [19], revealing their competitive relationship with the temperature and pressure. Besides, we also include another pathway (R6) for $\text{CH}_3\text{OCH}_2 + \text{O}_2$ proposed by Suzuki et al. [18], directly forming the HCO radical, H_2O and formaldehyde, which may also increase the agreement between measurements and predictions at lower pressures.



The methoxymethyl peroxy radical ($\text{CH}_3\text{OCH}_2\text{OO}$) undergoes two further kinds of reactions, i.e. isomerization via the intramolecular rearrangement of the peroxy radical (R7), and its self-reactions (R8) and (R9):



The rate constants of R7–R9 used in the Zhao model [7] and in the literature are presented in Fig. 6a and b. In the updated sub-mechanism, the data for R7 by Suzuki et al. [17] was adopted. For R8 and R9, as can be seen from Fig. 6b, the total rate constants by Rosado-Reyes et al. [19] are consistent with the values obtained by Jenkin et al. [63] at 298 K and present a weak negative temperature dependence, while the values by the Zhao model are far

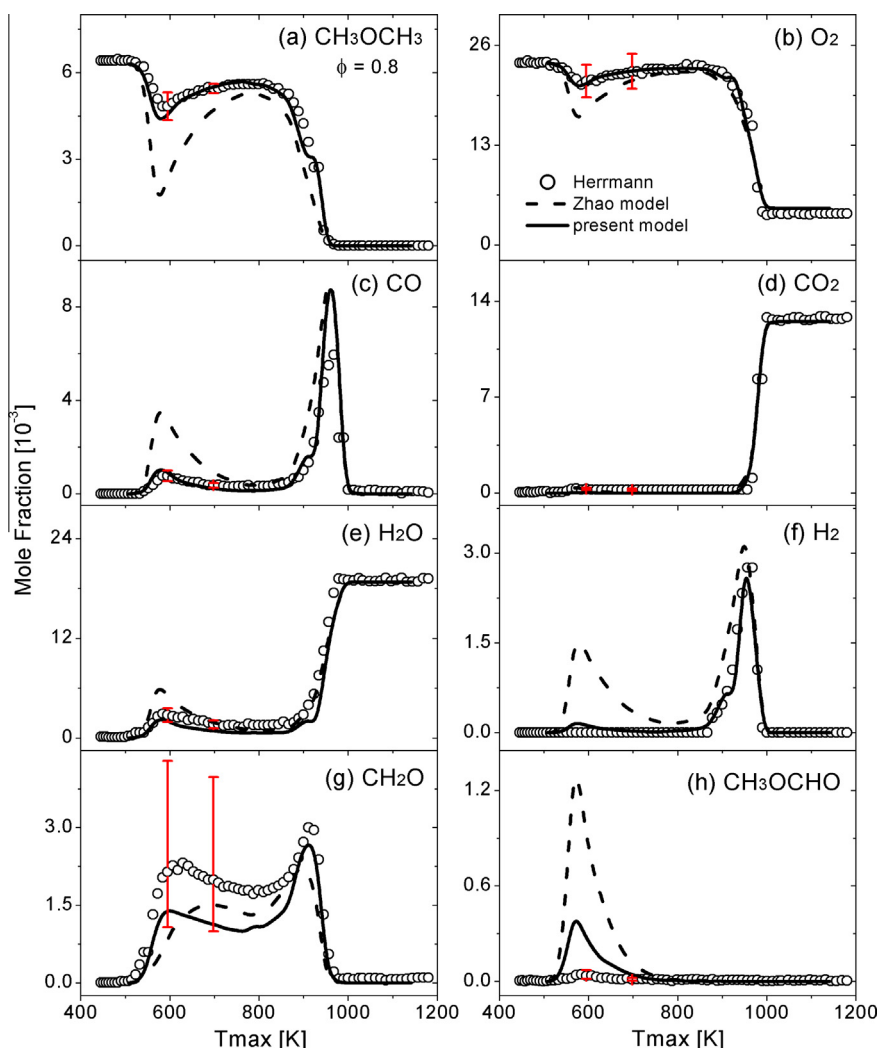


Fig. 9. Mole fraction profiles of selected species as a function of temperature in lean DME oxidation ($\phi = 0.8$). Symbols are experimental results from Herrmann et al. [22], lines are simulation results. Dashed lines: Zhao model [7]; solid lines: present model.

away, especially with increasing temperature. In this work, the total rate constants of R8 and R9 by Rosado-Reyes et al. [19] were adopted in the updated sub-mechanism, and their branching ratio was estimated according to the work of Jenkin et al. [63].

The reactions of the second O₂ addition to CH₂OCH₂OOH (R10, see below) and the decomposition of CH₂OCH₂OOH (R11) were then investigated (Fig. 6c and d). Although R10 and R11 are very important in DME low-temperature oxidation, available data are scarce, especially for R10. For R11, the rate constants by Yamada et al. [39], Suzaki et al. [17] and the Zhao model [7] are consistent (Fig. 6d). Andersen and Carter [38] also investigated the rate constants of R11 at 1, 10 and 40 atm as well as at the high-pressure limit. A significant pressure-dependent effect was observed at temperatures >400 K. However, the HPL rate constants are much higher than those from Yamada et al. [39], Suzaki et al. [17] and the Zhao model [7], as shown in Fig. 6d. Considering the large discrepancies between the results by Andersen and Carter and other data, we kept the rate constants of R10 and R11 as the same as in the Zhao model in the updated sub-mechanism.



The subsequent pathways of OCH₂OCH₂OOH → HOOCH₂OCHO → OCH₂OCHO through R12 and R13 were also used as in the Zhao model [7], while the following decomposition pathways of OCH₂OCHO were adopted from the present calculations, which have been discussed in detail in the theoretical calculation section. All updated rate constants used in this work, as well as some other important ones are listed in Table 2.

The thermodynamic data are mainly kept from the Zhao model [7], with the newly defined species OCHOCHO calculated in this work. The detailed calculation method for deriving its thermodynamic data can be found in the Supplementary Material.

4.2. Simulation methods

The simulation was performed using the plug-flow reactor module of the Chemkin-Pro software [64]. The problem type was selected as “fix gas temperature”, and the measured and interpolated temperature profiles were used as input with T_{max} steps of 20 K in the simulation. The pressure inside the reactor was set as constant at 1 atm and the volumetric flow rate was set as 0.5 SLM (std. conditions: 273.15 K, 1 atm). In the calculation, the Solver Maximum Step Distance was set as 0.01 cm. Since the composition with the reaction process was sampled at the outlet of the reactor, the mole fractions at $L = 830$ mm with each temperature profile were used as simulation results.

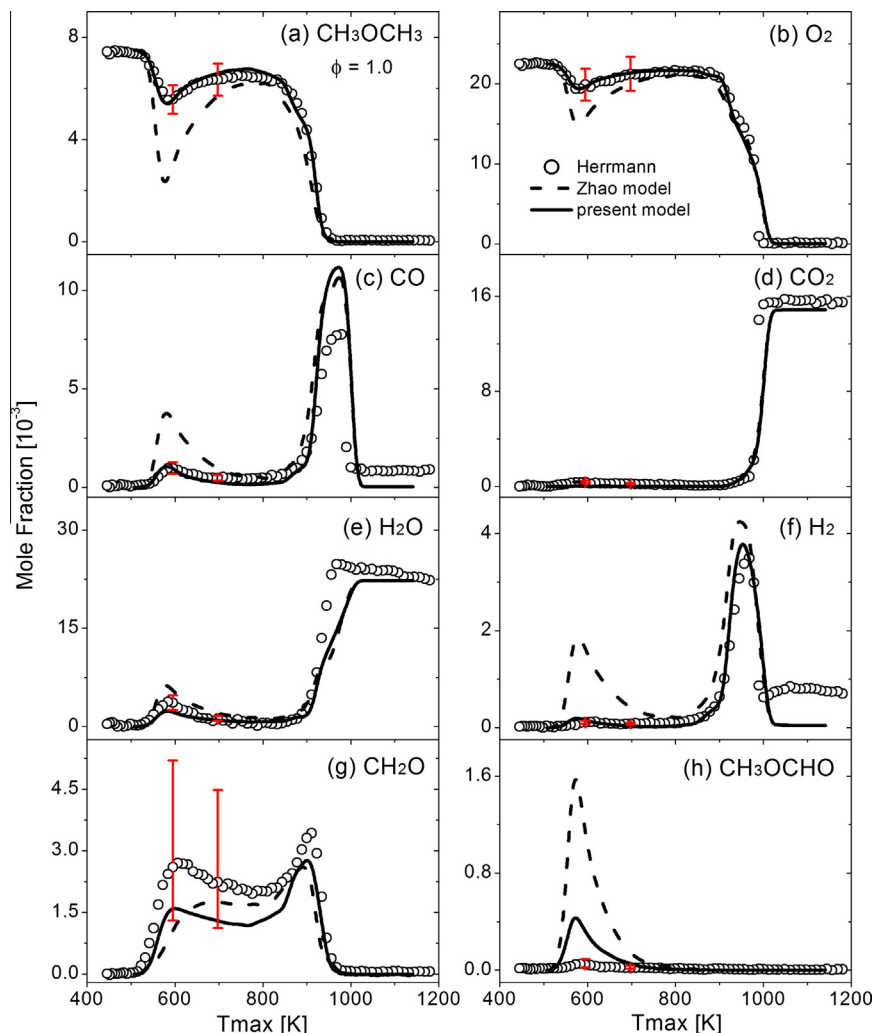


Fig. 10. Mole fraction profiles of selected species as a function of temperature in stoichiometric DME oxidation ($\phi = 1.0$). Symbols are experimental results from Herrmann et al. [22], lines are simulation results. Dashed lines: Zhao model [7]; solid lines: present model.

5. Results and discussion

During the oxidation of DME at stoichiometric condition, a large number of species were identified. Reactants and products including formaldehyde (CH_2O), H_2 , CO , and H_2O show two-stage ignition behavior. Methyl formate (CH_3OCHO) was also recorded as an important species at low temperature. These species can provide a stringent examination for our proposed chemical kinetic mechanism of DME, especially in the low-temperature regime. In addition, we observed a species with $m/z = 64$, which could not be identified exactly, and further discussions on this species can be found in the Supplementary Material (Fig. S1 and Table S1).

Figure 7 shows the comparison of the measurements and the predictions by the present model and the Zhao model. As we can see, the Zhao model [7] predicts much higher oxidation reactivity than our measurement in the low- and intermediate-temperature regions, while the present model prediction reproduces the experimental observations very well. Thus the following discussions are focused on the improvement of the present model regarding the oxidation reactivity of DME in the low- and intermediate-temperature regimes, compared with the Zhao model [7]. In addition, the reaction characteristics of the first-stage ignition and the negative temperature coefficient (NTC) zone are revealed simultaneously by the reaction flux analysis at 593 and 695 K, respectively, to better understand the low-temperature oxidation process, as can be seen

in Fig. 8. The reaction pathway analysis at 593 K based on the Zhao model is shown as Fig. S2 in the Supplementary Material.

5.1. DME oxidation reactivity in low- and intermediate-temperature regimes

From Fig. 7 it is seen that large discrepancies exist between our measurement and the prediction by the Zhao model [7] in the low- and intermediate-temperature regimes, especially at 593 K. The major modifications of the present model in the DME sub-mechanism are discussed in the kinetic modeling part. In this section, only the updated pathways or rate constants that play significant roles in decreasing the oxidation reactivity of DME are discussed, helping us to better understand the low-temperature oxidation process.

Following the reaction sequences, we firstly find that the improvement of the first O_2 addition reactions ($\text{CH}_3\text{OCH}_2 + \text{O}_2$) is important in the fuel conversion and also in the formation of CH_2O . It is known that chemically-activated reactions play an important role in competing with thermally-activated reactions, especially at relatively low pressure and high temperature. In the present model, the experimental work by Rosado-Reyes et al. [19] was adopted for the first O_2 addition reactions, considering the competition between R4 and the chemically-activated reaction R5. As shown in Fig. 5d, R4 is the dominant channel at the

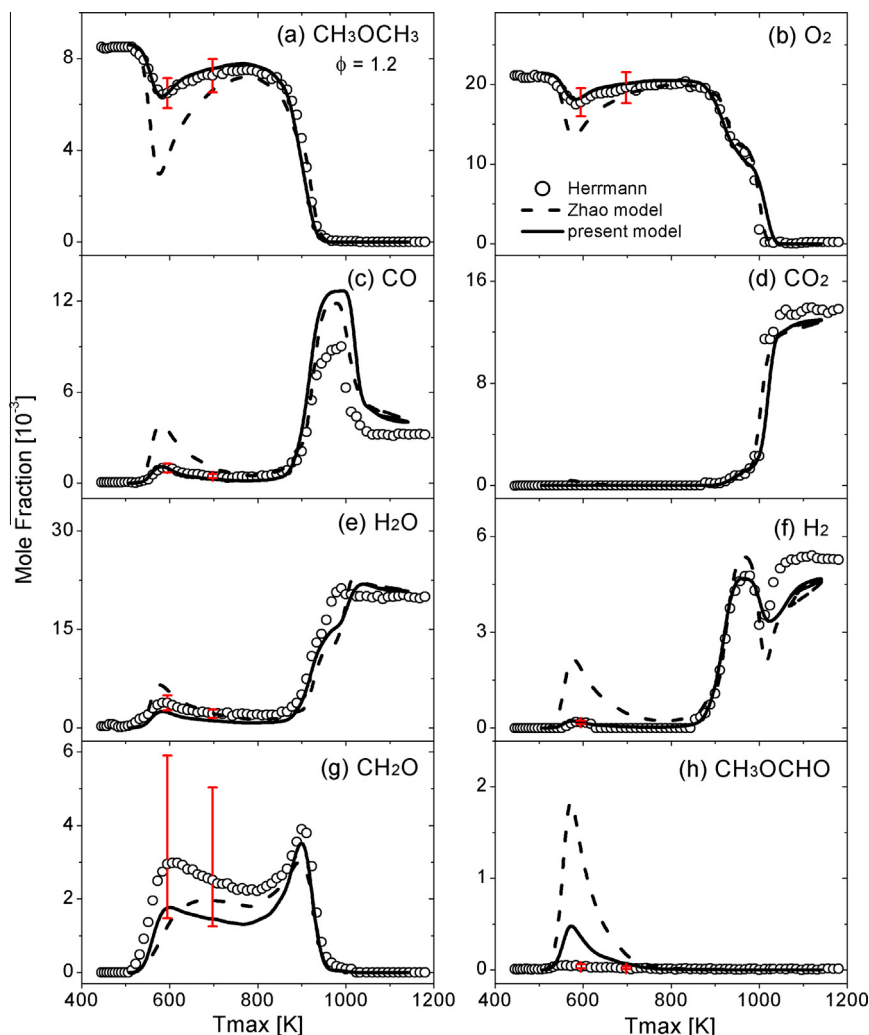


Fig. 11. Mole fraction profiles of selected species as a function of temperature in rich DME oxidation ($\phi = 1.2$). Symbols are experimental results from Herrmann et al. [22], lines are simulation results. Dashed lines: Zhao model [7]; solid lines: present model.

high-pressure limit, while R5 becomes competitive at 1 atm. Therefore, as seen in Fig. 8, under our experimental condition of atmospheric pressure, the reaction pathway analysis at 593 K shows that there is an additional branch from CH_3OCH_2 , leading to the formation of CH_2O and OH , consuming 19% of the reaction flux (highlighted by a box in Fig. 8a). As a result, the reaction flux reduction of the chain-branching reaction leads to lower oxidation reactivity, and the new pathway of R5 also promotes the early formation of CH_2O . The better prediction of CH_2O is also caused by the improvement of its two dominant consumption pathways, i.e. R23 and R24:



The calculated values by Xu et al. [65] for R23 and R24 were adopted in the present model, which are about 40% and two orders of magnitude lower, respectively, than those used in the Zhao model [7] in the 500–1000 K range. The decrease of the CH_2O consumption rates further increases its mole fraction.

Another significant contribution to decrease the oxidation reactivity of DME at relatively low temperature is the better description of the OCH_2OCHO decomposition mechanism through our high-level quantum chemistry calculations. In the Zhao model

[7], the dominant reaction sequence regarding this radical at 593 K is as follows: $\text{OCH}_2\text{OCHO} \rightarrow \text{HOCH}_2\text{OCO} \rightarrow \text{HOCH}_2\text{O} + \text{CO} \rightarrow \text{HOCHO} + \text{H} + \text{CO}$, as shown in Fig. S2 in the Supplementary Material. This reaction sequence has been proposed by Curran et al. [15] and was adopted by all previous models [5,7,15,25,40]. Therefore, in the predictions of the previous models, such as the Zhao model [7], large amounts of H atoms are produced. These H atoms further accelerate the consumption of DME by H-abstraction reactions, leading to the under-estimation of the reactants and the over-estimation of H_2 and other species (CO , H_2O , and CH_3OCHO) in the low-temperature region (see Fig. 7). According to our theoretical calculations, the rate constant of R14 is substantially higher than R15, as seen in Fig. 4, revealing that OCH_2OCHO favors forming HOCHO and HCO rather than HOCH_2OCO . Therefore in our proposed model, the dominant reaction sequence of OCH_2OCHO at 593 K, shown in Fig. 8a, is $\text{OCH}_2\text{OCHO} \rightarrow \text{HOCHO} + \text{HCO} \rightarrow \text{HOCHO} + \text{CO} + \text{HO}_2$. Compared with the Zhao model [7] prediction, the HO_2 radical rather than the H atom is produced eventually in the present model. The large production of HO_2 promotes the formation of H_2O_2 through the self-recombination reaction ($\text{HO}_2 + \text{HO}_2 = \text{H}_2\text{O}_2 + \text{O}_2$), resulting in chain termination. Thus in the present model prediction, the oxidation reactivity is largely decreased compared with the Zhao model [7], which is seen especially in the significant improvement of DME and H_2 mole fractions

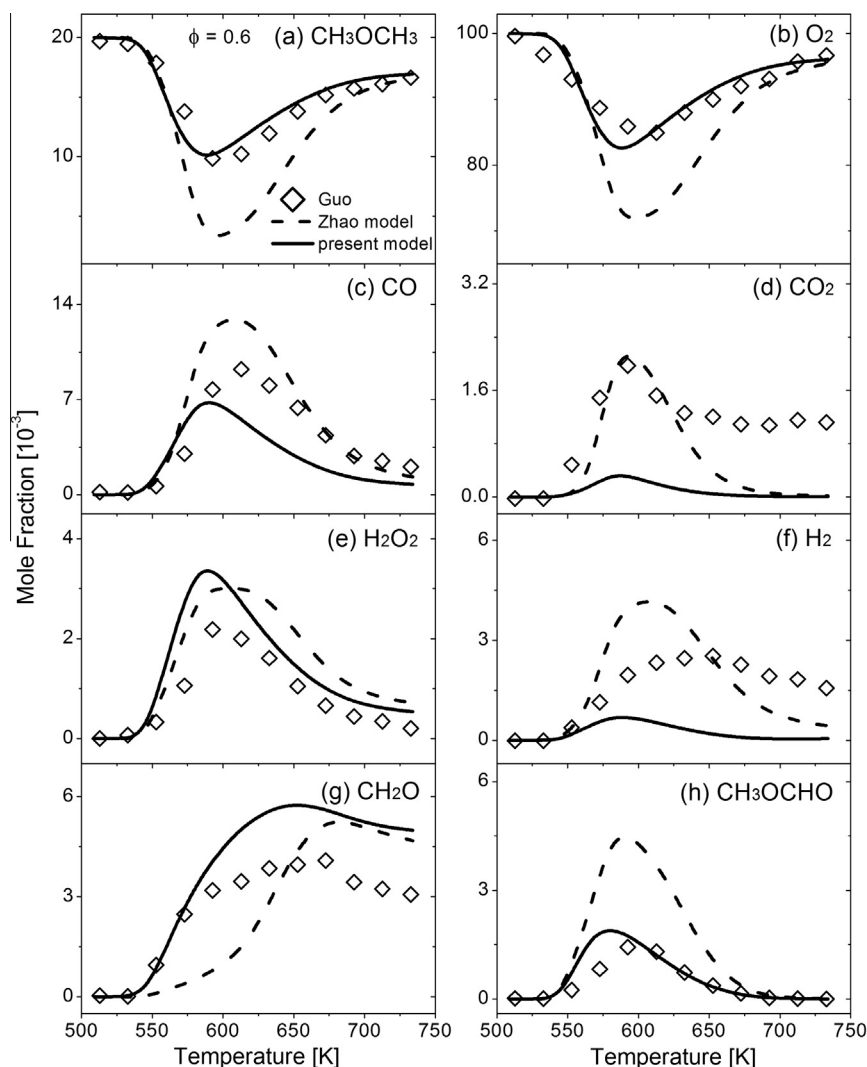


Fig. 12. Mole fraction profiles of selected species as a function of temperature in lean DME oxidation ($\phi = 0.6$). Symbols are experimental results from Guo et al. [21], lines are simulation results. Dashed lines: Zhao model [7]; solid lines: present model.

(see Fig. 7). The change of the radical pool can affect the fuel conversion in both the low- and intermediate-temperature regions, and our prediction can thus capture the oxidation reactivity of DME quite reliably.

The present model was further examined against recent flow reactor oxidation experiments at atmospheric pressure by Herrmann et al. [22], Guo et al. [21] (see Figs. 9–12). Moreover, it was also compared against earlier experimental datasets at higher pressure, including flow-reactor oxidation at 12.5 atm with an equivalence ratio of 1.19 measured by Curran et al. [15], JSR oxidation data at 10 atm with equivalence ratios of 0.2 and 1.0 measured by Dagaut et al. [14], and ignition delay times at 12.8 and 39.5 atm with equivalence ratio of 1.0 measured by Pfahl et al. [13]. The comparisons of the model prediction with the experimental results are given in the [Supplementary Material \(Figs. S3–S6\)](#). A generally good agreement was found between the modeling and these data in the literature. Thus, the developed sub-mechanism of DME low-temperature oxidation in this work is suitable to simulate DME oxidation at 1 atm and also at higher pressures (10–40 atm).

Additionally, the recent work by Kurimoto et al. [24] modified the branching ratio of $\text{CH}_2\text{OCH}_2\text{OOH}$ decomposing to CH_2O (R11) by increasing the A factor of R11 to $5.3224 \times 10^{13} \text{ s}^{-1}$, which is a factor of 3.5 higher than that used in the Zhao model. The increase of the rate constant of R11 directly affects the chain branching and reduces the DME low-temperature oxidation reactivity [24]. We also tried to simulate the experimental data by adopting the recommended value of R11 by Kurimoto et al. in the updated sub-mechanism, which presents a much lower reaction activity compared with the experimental measurement in this work, and also in the literature [13–15,21,22]. The experimental data by Kurimoto et al. [24] were used to validate the present model, as well as the Zhao model (see Fig. S7 in [Supplementary Material](#)).

5.2. The comparison of reaction characteristics between the first-stage ignition and the NTC zone

Seen from our measurements in Fig. 7, in the temperature region of the first-stage ignition (around 593 K), reactants were partially consumed to form some key species including H_2 , CO, H_2O , CH_2O and CH_3OCHO . As the temperature increases to the NTC zone, the consumption of reactants and the formation of intermediates decrease. To better understand the auto-ignition process, reaction pathway analyses at 593 and 695 K were performed, to represent the characteristics of the first-stage ignition and the NTC zone, respectively. By comparing Fig. 8a and b, notable differences can be observed at the two temperatures. Firstly, with increasing temperature, the chemically-activated reaction R5 becomes more competitive, contributing 31% at 695 K compared to 19% at 593 K, which further reduces the oxidation reactivity. Secondly, the $\text{CH}_3\text{OCH}_2\text{OO}$ radical dominantly isomerizes to the $\text{CH}_2\text{OCH}_2\text{OOH}$ radical (R7), while the channels to methyl formate through R8 and R9 become negligible at 695 K. Most importantly, the branching ratio of the $\text{CH}_2\text{OCH}_2\text{OOH}$ radical reactions is changed greatly. At 593 K, the $\text{CH}_2\text{OCH}_2\text{OOH}$ radical mainly reacts through the second O_2 addition reaction (R10), and then produces two OH radicals. However, the decomposition channel of this radical via R11 becomes dominant at 695 K, which results in the formation of CH_2O . More significantly, the lower production of OH radicals largely decreases the reactivity of DME in the NTC zone.

6. Conclusions

In this work, a combination of a laminar flow reactor experiment with EI-MBMS species detection was used to investigate the low- and intermediate-temperature oxidation of DME at highly

argon-diluted stoichiometric conditions and atmospheric pressure, which provides a target for the further development of a DME oxidation model. The rate constants of reactions affecting the oxidation reactivity of DME at relatively low temperatures were reviewed with care in light of recent experimental and theoretical results. The reaction pathways of the OCH_2OCHO radical were investigated in detail by high-level quantum chemistry calculations. A new pathway through β -scission accompanied by H-transfer of OCH_2OCHO was established, which dominantly leads to the formation of formic acid and HCO in the temperature range from 500 to 800 K.

An updated sub-mechanism of DME oxidation was developed via the rate constant evaluation and theoretical calculations in order to explain the oxidation reactivity of DME, especially in the low- and intermediate-temperature regimes. The chemically-activated pathway of $\text{CH}_3\text{OCH}_2 + \text{O}_2$, which directly leads to the formation of formaldehyde, has a large effect on the DME low-temperature oxidation reactivity at 1 atm. The radical pool shifting from the reactive H atom to the unreactive HO_2 radical is another reason leading to the reduction of the low-temperature reactivity. The model improvement was examined against the experimental data from this work, and also against datasets reported in the literature, covering a wide range of pressure, temperature and equivalence ratios. The goal of this work is to provide a preliminary assessment of the dominant reactions affecting the DME low-temperature chemistry, which can be used to guide the model development in the future. Comprehensive and systematic theoretical investigations on the DME low-temperature oxidation network are desired to develop an accurate DME oxidation mechanism.

Acknowledgments

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Appendix A. Supplementary material

1. Experimental methods, observation of the species at $m/z = 64$, theoretical method for deriving the thermodynamic data of OCHOCHO and model comparison with data in the literature. 2. Mechanism, thermochemical and transport data in Chemkin format. 3. Species information with geometries, energies and frequencies calculated in this work. Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2014.10.003>.

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